

Computer Science Division

Machine Learning
CS441/741

Assignment 4: For this assignment, you have a choice of either exploring ensemble learning or unsupervised learning.

Due: 19 October 2026, 08:00

Instructions

For the purposes of this assignment, please note the following:

- You may select any one of the options listed below.
- Please submit your own work.
- You may program in any language, provided that your code compiles and execute on Linux. While you are allowed to make use of machine learning libraries, note that you learn most about the machine learning algorithms when you implement these algorithms yourself.
- You have to write a report, using the IEEE conference template (google!), in two-column, 10pt format. Please see assignment 2 for tips on report writing. Also see the writing rules – uploaded to your STEMlearn module – for a list of writing rules. Please consult these writing rules and apply them. Note that only the report will be evaluated, not your code. Therefore, a bad report will result in bad marks.
- Submit your report in pdf format. Note that no format other than pdf will be accepted. Provide a link to the Git repository that contains your code. Code will not be evaluated, but may be scrutinized if found necessary. Make sure that you name your report file as `????????RWxxxassignment3.pdf`, where you replace the question marks with your student number, and `xxx` with the module code that you are

registered for. Please note that I use a script to pull out all reports, and if you do not follow this file naming convention, your report **will not** be extracted for evaluation.

- Not that the report must be a formal technical report wherein you report all that you have done, your results, and a discussion of these results. For guidelines on the format of a technical report and a mark rubric, see the last sections.
- You **must** include your student number in your report title section. You **must not** include your name, surname, or other similarly identifying information in your report title section.
- Generative AI tools may not be used.
- Upload your report via STEMlearn before the deadline of **19 October 2026, 08:00**. Note that late assignments will not be accepted. After this deadline, I will extract all reports to start with evaluation of the reports that day.

Option 1: Ensemble learning

The main objective of this assignment is to explore ensemble learning as a means of predicting cyberattacks. Reusing the dataset provided for assignment 2, select an appropriate ensemble method and tune this method accordingly. Ensure that the choice of ensemble method and the tuning performed are well justified. Any necessary data transformations should also be performed where well justified. Evaluate the tuned ensemble method on the dataset and compare the results with the models used in assignment 2.

You may select any ensemble learning model, but approach this decision with intention. What do you wish the final evaluation to reflect? Do you want to compare ensemble methods that are similar to the models used in assignment 2, or do you wish to want to compare dissimilar methods? Neither option is correct, but a sound argument must be supplied.

The tuning of your chosen ensemble method should also be performed with an aim to answer specific questions that are set out before evaluation. Not only do you want a model that performs well, but you also want to gain insight from the tuning performance.

You report should include the necessary background, detail on your chosen ensemble method and the implementation thereof, detail on the empirical process followed and then finally your results and a discussion of the results. Remember that you must include all necessary background and implementation details for the models used in assignment 2 that you compare against. You may revise or improve your assignment 2 implementation.

Option 2: Unsupervised learning using Self-Organising Maps

The main objective of this assignment is to explore unsupervised learning through self-organising maps (SOM) as a means of predicting cyberattacks. Reuse the dataset provided for assignment 2 with the target feature removed. Make sure to apply the necessary data transformations, and to tune the SOM. Train a SOM on the edited dataset and cluster the trained codebook vectors. Devise an approach to label each cluster with a target cyberattack label. Evaluate the ability of your SOM model to predict the target class of instances in a generalization set. Compare the results with the models used in assignment 2.

The training of the SOM should be very well documented and optimisation procedures should be well justified. Choice regarding target class assignment should be well justified or tested.

Your report should include the necessary background, detail on SOMs in general and your implementation, detail on the empirical process followed and then finally your results and a discussion of the results. Remember that you must include all necessary background and implementation details for the models used in assignment 2 that you compare against. You may revise or improve your assignment 2 implementation.

Mark Rubric

Your report will be evaluated as follows:

Aspect	**Mark**
Abstract	4
Introduction	8
Background	24
Implementation	8
Empirical process	18
Results & discussion	48
Conclusions	4
References	4
Linguistic quality	32
Total	150